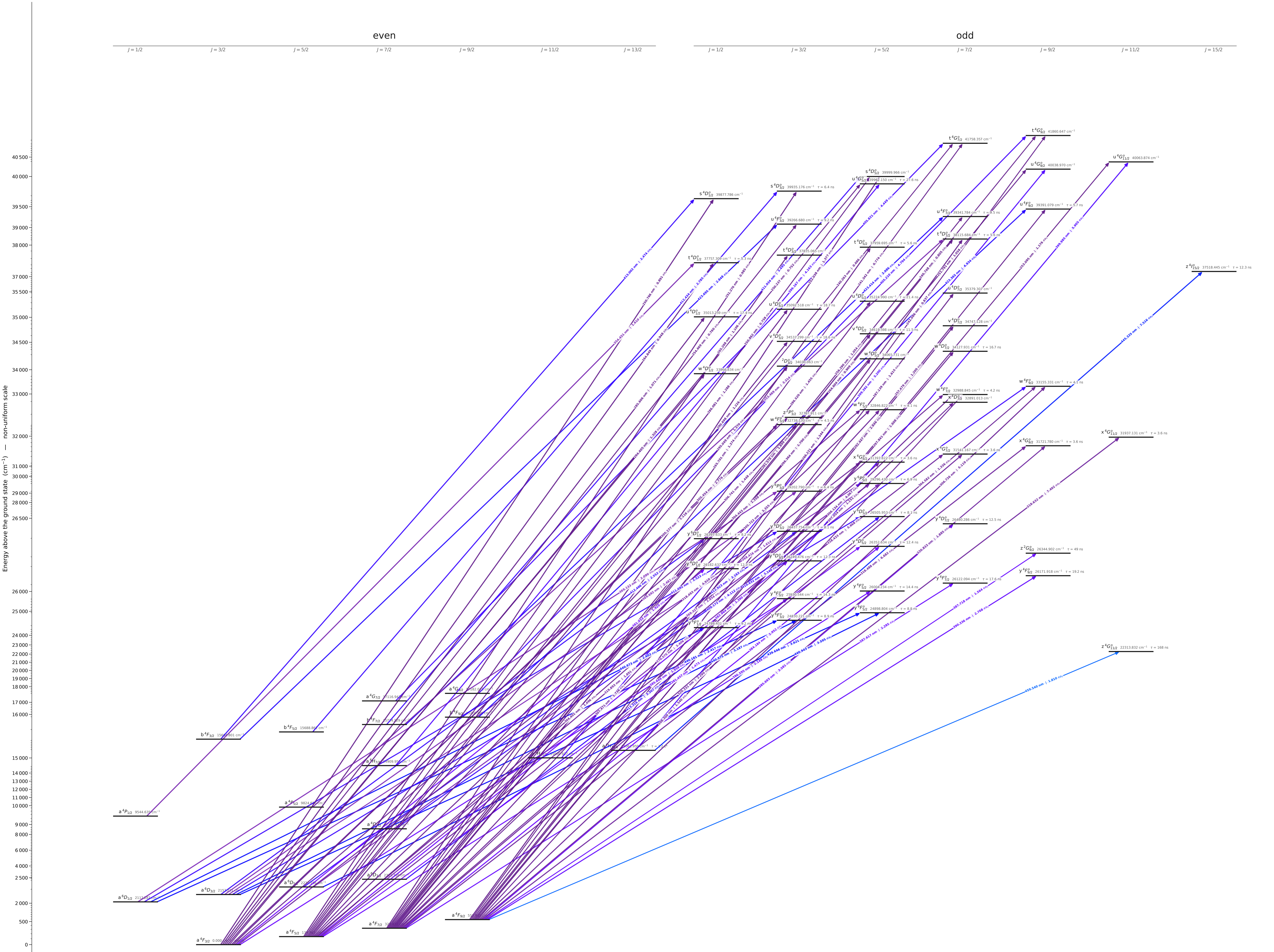


V⁺ ionisation limit 54411.67 cm⁻¹ = 6.74619 eV (183.784 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle f|er|f'\rangle|$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists)



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ U e J J $ (ea0)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$a^4H_{132} - z^4F_{15Q}$	445.3255	445.2005	673.1985	7.516	8.100e+07	12.94	100	laser-cooling

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
y^0_{F5}	9930	-	F=1: -1112.98 F=2: -915.12 F=3: -618.32 F=4: -222.60 F=5: +272.06 F=6: +865.65
$y^1_{D5}x^0_{F5}$	1530	-	F=1: -168.63 F=2: -138.65 F=3: -93.69 F=4: -33.73 F=5: +41.22 F=6: +131.16
y^1_{F5}	54060	-	F=2: -3642.48 F=3: -2023.60 F=4: +134.91 F=5: +2833.04
$y^2_{F5}x^0_{F5}$	39060	-	F=3: -4381.48 F=4: -3505.09 F=5: -2346.81 F=6: +876.79 F=7: +1071.76 F=8: +3410.14
$x^1_{F5}x^0_{F5}$	60310	-	F=1: -6779.06 F=2: -5733.89 F=3: -3765.14 F=4: -1355.81 F=5: +1657.10 F=6: +5272.60
$x^2_{F5}x^0_{F5}$	10000	-	F=2: -4580.08 F=3: -3707.69 F=4: -2579.29 F=5: -1677.09 F=6: +367.09 F=7: +1526.69 F=8: +3562.29
$x^2_{F5}x^1_{F5}$	15900	-	F=1: -3058.63 F=2: -2740.65 F=3: -264.18 F=4: -1628.62 F=5: +83.14 F=6: -119.17 F=7: +1231.40 F=8: +2502.52
$x^3_{F5}x^1_{F5}$	81100	-	F=2: -1841.48 F=3: -1598.64 F=4: -1274.87 F=5: +870.75 F=6: -384.48 F=7: +182.12 F=8: +829.68 F=9: +1558.17
$w^0_{F5}x^0_{F5}$	25800	-	F=1: -2900.50 F=2: -2384.85 F=3: -1611.34 F=4: -580.10 F=5: +709.01 F=6: +2255.94
$w^1_{F5}x^0_{F5}$	16900	-	F=2: -1699.82 F=3: -1591.89 F=4: -1376.04 F=5: -1052.27 F=6: -620.57 F=7: -60.94 F=8: +566.61 F=9: +1322.08
$z^0_{F5}x^0_{F5}$	18130	-	F=4: -5440.47 F=5: -4526.11 F=6: -3428.87 F=7: -2148.76 F=8: -685.77 F=9: +960.08 F=10: +2788.81 F=11: +4800.42
$z^1_{F5}x^0_{F5}$	21310	-	F=2: -1436.76 F=3: -798.20 F=4: +53.21 F=5: +1117.48
$t^0_{D5}x^0_{F5}$	14430	-	F=0: -2266.43 F=1: -2122.53 F=2: -1834.72 F=3: -1403.03 F=4: -827.42 F=5: -107.92 F=6: +755.48 F=7: +1467.85
$u^1_{F5}x^0_{F5}$	28510	-	F=2: -5482.46 F=3: -4912.85 F=4: -4058.44 F=5: -2919.23 F=6: -1495.22 F=7: +213.60 F=8: +2207.22 F=9: +7742.78

Measured A, B constants; sources in the data file